

Order of Operations (GEMDAS)

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS GEMDAS: **G**rouping first, then **E**xponents, then **M**ultiply and **D**ivide from left to right, then **A**dd and **S**ubtract from left to right. Worked example: $3 + 2 \times (5 - 1)^2 = 3 + 2 \times 16 = 35$.

Round 1 — Retrieve It

1. What does each letter in GEMDAS stand for?

2. Which step comes first, grouping or exponents?

3. In what direction do you work multiply and divide?

4. Do add and subtract have to be done in that exact order, or left to right?

Round 2 — Sketch It

Rewrite $3 + 2 \times (5 - 1)^2$ step by step down the page. Circle the operation you complete at each step.

Order of Operations (GEMDAS) — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Evaluate $4 + 6 \div 2$.

2. Evaluate $(8 - 3) \times 2^2$.

3. Evaluate $10 - 2 \times 3 + 1$.

Round 4 — Reason from Evidence

A student evaluates $3 + 2 \times (5 - 1)^2$ and gets $3 + 2 \times 8 = 40$. Find the mistake, explain which GEMDAS step went wrong, and give the correct answer.

Order of Operations (GEMDAS) — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain why order of operations matters. Use one example that is not printed on the cue card to show how a different order changes the answer.

C

Connect

Connect the "left to right" rule for multiply/divide to the same rule for add/subtract. Why do we need a direction rule when operations share the same level?

E

Extend

Create your own expression that uses grouping and an exponent. Show each GEMDAS step down to the final value.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Order of Operations (GEMDAS)

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Grouping, Exponents, Multiply/Divide, Add/Subtract.
2. Grouping.
3. Left to right.
4. Left to right (not always add before subtract).

Sketch It — look for

Steps shown in order: $(5 - 1) = 4 \rightarrow 4^2 = 16 \rightarrow 2 \times 16 = 32 \rightarrow 3 + 32 = \mathbf{35}$, with the current operation circled at each step.

Apply It

1. $4 + 6 \div 2 = 4 + 3 = \mathbf{7}$.
2. $(8 - 3) \times 2^2 = 5 \times 4 = \mathbf{20}$.
3. $10 - 2 \times 3 + 1 = 10 - 6 + 1 = \mathbf{5}$.

Reason from Evidence — look for

The exponent step was done wrong: $(5 - 1)^2 = 4^2 = 16$, not 8. Correct: $3 + 2 \times 16 = 3 + 32 = \mathbf{35}$.

ACE — Extend look-for

Many answers possible. Check that the expression includes grouping and an exponent and that each GEMDAS step is shown correctly to a final value.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.